

Program 7 : Deploy Web Application to Kubernetes using Minikube

file:

|-----nginx-deployment.yaml

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$
```

Steps:

1) minikube start

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ minikube start
minikube v1.37.0 on Ubuntu 22.04
  KUBECONFIG=/home/1RV24MC108_SWASTHIK_DEVADIGA/.kube/config
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
docker "minikube" container is missing, will recreate.
Creating docker container (CPUs=2, Memory=3072MB) ...

Docker is nearly out of disk space, which may cause deployments to fail! (98% of capacity). You can pass '--force' to skip this check.
Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
Related issue: https://github.com/kubernetes/minikube/issues/9024

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Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Using image docker.io/kubernetes/dashboard:v2.7.0
  Using image docker.io/kubernetes/metrics-scraper:v1.0.8
```

2) apply nginx-deployment.yaml and check status and expose the deployment service

```
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           19h
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5lqf7   1/1     Running   2 (66s ago)  33m
nginx-deployment-6f9664446b-d5nqw   1/1     Running   2 (66s ago)  33m
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          23d
kubia               NodePort    10.106.174.127 <none>        8080:32481/TCP   23d
nginx-deployment    NodePort    10.110.47.236 <none>        80:30331/TCP     11s
nginx-service       NodePort    10.97.155.196 <none>        80:30866/TCP     19h
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30331

Opening service default/nginx-deployment in default browser...



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

